

A 2015 Data Analysis of Crime, Economics, and Policing in Austin, Texas

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CS5830: Data Science in Practice

Introduction

Crime affects how safe and healthy a city feels, so it is important for city planners, police leadership, and residents to understand what may be connected to higher crime and how often reported crimes are solved. This project analyzes crime reports from Austin, Texas, in 2015 and compares them with information about unemployment, housing, and public transit in different zip codes. We focused on four main questions: whether areas with higher unemployment also have more crime, whether better access to public transit is linked to more auto theft, whether more crimes happen on weekdays or weekends, and whether severe crimes are more likely to be cleared by police.

The results showed that areas with higher unemployment generally had higher crime rates. Areas with better access to public transit also tended to have more auto theft. More crimes were reported on weekdays than on weekends, and severe crimes were usually cleared more often than less severe crimes, although there was one exception. These findings can help city planners and police departments better understand crime patterns and make more informed decisions. Please find the [presentation slides](#) and [project repository](#) linked here.

Dataset

We used two datasets covering Austin, Texas, in 2015. The first contains 38,573 crime incidents, with information on offense type, zip code, report date, clearance status, and neighborhood characteristics such as unemployment and access to public transit. The second provides zip-code population data, which we used to calculate crime rates per capita instead of relying only on raw crime counts. These attributes make the datasets well suited for analyzing crime and public safety because they allow us to examine how crime patterns relate to neighborhood conditions, when crimes occur, and whether reported cases are ultimately cleared by police.

Analysis Techniques

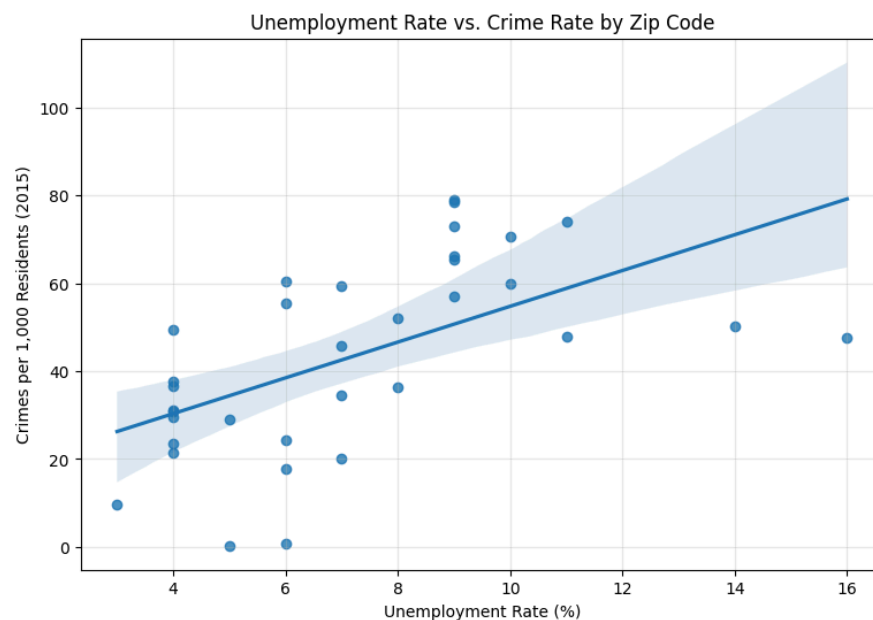
For our four questions, we used three statistical techniques. Pearson correlation was used to examine the relationship between unemployment and crime rate, transit access and auto theft, and crime severity and clearance, since each of these involves two continuous variables and we wanted to measure the strength and direction of their relationship without assuming one causes the other. For the transit analysis, we also used a Mann-Whitney U test to compare auto theft rates between high- and low-transit zip codes, since auto theft rates were skewed rather than normally distributed, making a rank-based test more appropriate than a t-test. Finally, we used an independent-samples t-test to compare average daily crime counts on weekdays and weekends, which was appropriate here because daily crime counts across a full year are numerous enough

to be approximately normally distributed. Together, these methods let us match each statistical test to the type of comparison and the shape of the data behind each question.

Results

Does the unemployment rate relate to the crime rate?

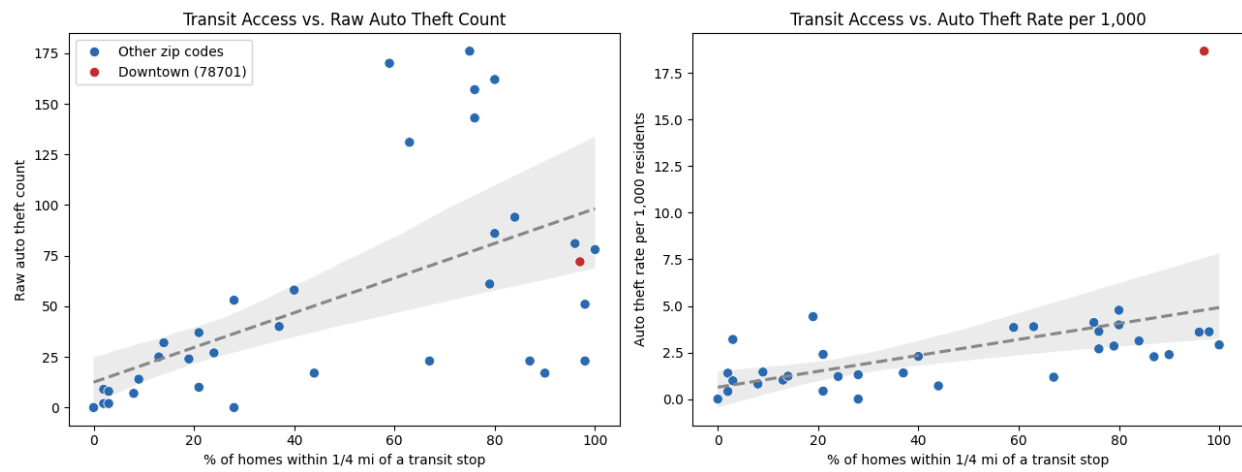
The zip codes with higher unemployment generally had higher crime rates. After excluding downtown Austin (ZIP code 78701), which was an extreme outlier, the relationship was moderately strong and statistically significant ($r = 0.571$, $p < 0.001$). Downtown had an unusually high crime rate of about 545 crimes per 1,000 residents, likely because relatively few people live there compared with the large number who travel there for work, shopping, and entertainment. When downtown was included, the relationship became much weaker and was no longer statistically significant ($r = 0.236$). This shows that the correlation analysis was useful for identifying an overall relationship, but it was also sensitive to unusual areas such as downtown. For workforce development programs and city leaders, unemployment may be one useful factor to consider when identifying communities that could benefit from additional economic resources, although the results do not show that unemployment itself causes crime.



Does transit access relate specifically to auto theft?

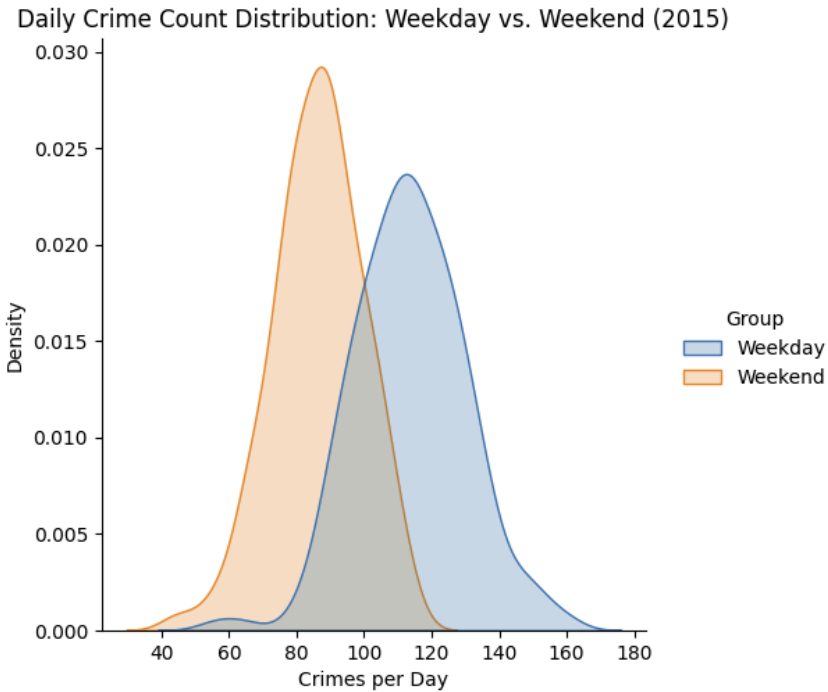
Zip codes with more homes near public transit generally had higher auto theft rates. This relationship appeared both when we looked at the total number of auto thefts ($r = 0.564$, $p < 0.001$) and when we adjusted for differences in population ($r = 0.492$, $p = 0.003$). The relationship became even stronger after excluding downtown Austin outlier ($r = 0.654$, $p < 0.0001$). A second statistical test showed the same pattern: high-transit zip codes averaged 3.99 auto thefts per 1,000 residents, compared with 1.41 in low-transit zip codes ($p < 0.001$). The fact that both analyses produced similar results strengthens the evidence that transit access and auto

theft are related. For CapMetro and city planners, this finding may help identify areas where additional vehicle-security measures, such as improved lighting, cameras, or secure parking, could be considered.



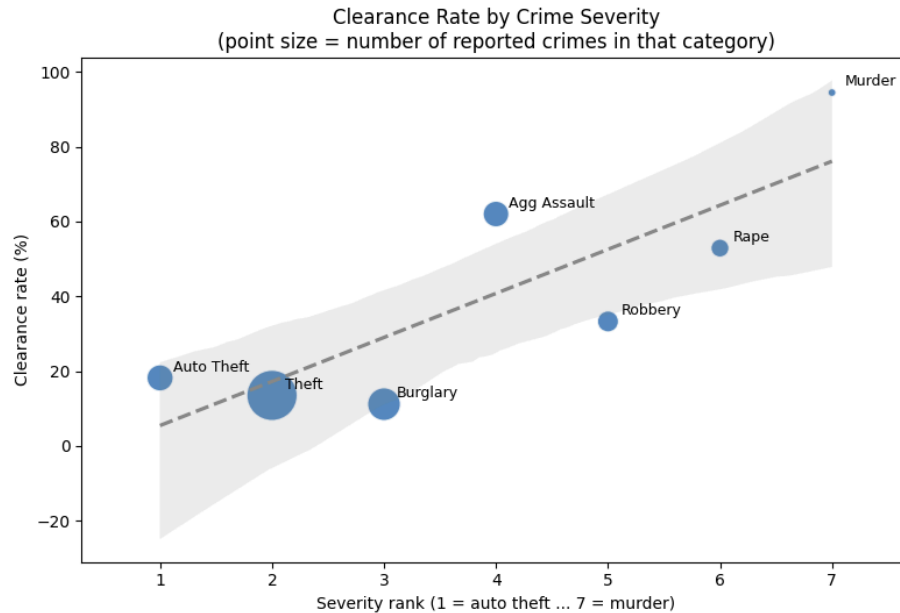
Does crime happen more on weekdays or weekends?

Weekdays averaged significantly more reported crimes per day than weekends, with about 113.4 crimes reported on a typical weekday compared with 86.2 on a weekend ($t = 15.13$, $p < 0.0001$). This may be unexpected if crime is assumed to peak on weekends. One possible explanation is that the dataset contains many property crimes, such as theft, burglary, and auto theft, which may be influenced by weekday work, travel, and other routine activities. For the Austin Police Department, this finding suggests that staffing decisions should consider the higher volume of reported crime occurring during the week rather than assuming that weekends always require the greatest coverage. The result shows that comparing average daily crime counts was useful for identifying a clear weekly pattern that could support patrol and staffing decisions.



Does crime severity relate to whether a case gets cleared?

More severe crimes were generally more likely to be cleared than less severe crimes. Across individual crime reports, the relationship was positive but relatively weak ($r = 0.214$, $p < 0.0001$, $n = 36,649$), while the pattern was much stronger when the seven offense categories were compared as a whole ($r = 0.828$, $p = 0.022$). One notable exception was robbery. Although it ranked as more severe than aggravated assault, only about 33% of robberies were cleared compared with 62% of aggravated assaults. This shows that severity is related to clearance, but severity alone does not explain whether a crime will be cleared. For Austin Police Department leadership, the unusually low robbery clearance rate compared with other serious crimes may be an area worth examining further to understand what factors make these cases more difficult to resolve.



Technical

For the data preparation, we subset each table to only the columns needed for our analyses to keep the processing. Percentage values for unemployment and transit access were converted from text to numeric values, and comma-formatted population figures were cleaned before use. Because neighborhood-level economic and housing characteristics were repeated for every crime within the same zip code, we reduced these data to one row per zip code before merging them with population data to calculate crime rates per-capita. Report dates were converted to date values so that day of the week could be identified. Records missing clearance status were removed, and crime severity scores were assigned to offense types using the FBI UCR Hierarchy Rule because the original dataset did not include a severity variable.

For the analysis and process, Pearson correlation was used to examine relationships involving unemployment, transit access, and crime severity, while a t-test was used to compare weekday and weekend crime counts. A Mann-Whitney U test was also used for the high- versus low-transit comparison because auto theft values were right-skewed. We also found that downtown Austin (ZIP code 78701) was an extreme outlier in the per-capita crime data. Rather than removing it without explanation, we repeated the unemployment and transit analyses both with and without this zip code and reported both results. Finally, because the meaning of the dataset's Clearance_Date field was unclear for some uncleared cases, the severity analysis relied only on Clearance_Status and did not use Clearance_Date.